

LECTURE:20 (Pretreatment)

➤ INTRODUCTION

- The past three decades have witnessed significant research & development on the front of production of biofuels and many other chemicals using renewable sources, such as lignocellulosic biomass (LCB).
- LCBs are evenly distributed on the globe and abundantly available in most of the countries.
- Further they are carbon neutral too, i.e. the CO₂ released by the products of LCBs, specially the biofuels is balanced by CO₂ absorption by the plants somewhere else. As a result the carbon footprint is minimized (A carbon footprint is the amount of greenhouse gases, particularly CO₂ produced by human activities).
- The majority of LCB is composed of biological polymers such as cellulose, hemicellulose, and lignin, which are strongly bound together by covalent and hydrogen bonds making the structure highly recalcitrant.
- The presence of lignin renders the bio-polymeric structure highly resistant to solubilization thereby inhibiting the hydrolysis of cellulose and hemicellulose which presents a significant challenge for the isolation of the respective bio-polymeric components.
- This has led to extensive research in the development of pretreatment techniques utilizing various physical, chemical, physicochemical, and biological approaches which are specifically tailored towards the source biomaterial and its application.
- The success of biofuel and chemical production strongly depends on the pretreatment methods used.
- Overall, pretreatment is the major step in the successful production of valuable products from LCBs.
- With the help of pretreatment, the bonds are broken and association amongst cellulose, hemicellulose and lignin is weakened for further application.
- In the present class note, a comprehensive discussion on various aspects of the pretreatment techniques, such as physical, chemical, thermophysical, thermochemical and biological methods have been made.
- For proper understanding of the topic, basics of hydrolysis has also been discussed.

➤ HYDROLYSIS

Basics

- In chemistry and physiology, hydrolysis is defined as a double decomposition reaction with water as one of the reactants.
- Thus, if a compound is represented by the formula AB in which A and B are atoms or groups and water is represented by the formula HOH , the hydrolysis reaction may be represented by the reversible chemical equation $AB + HOH \rightleftharpoons AH + BOH$.
- The reactants other than water, and the products of hydrolysis, may be neutral molecules—as in most hydrolyses involving organic compounds—or ionic molecules, as in hydrolyses of salts, acids, and bases.
- In other simple words, hydrolysis may be defined as a chemical reaction in which a substance reacts with water so as to be changed into one or more other substances, as

a starch into glucose, natural fats into glycerol and fatty acids, or a salt into a weak acid or a weak base.

- Here water is used to break down a compound; this is achieved by breaking a covalent bond in the compound by inserting a water molecule across the bond.
- The opposite of this is a dehydration-condensation reaction.
- There are commonly three types of hydrolysis reactions as depicted in **Figure-1**.

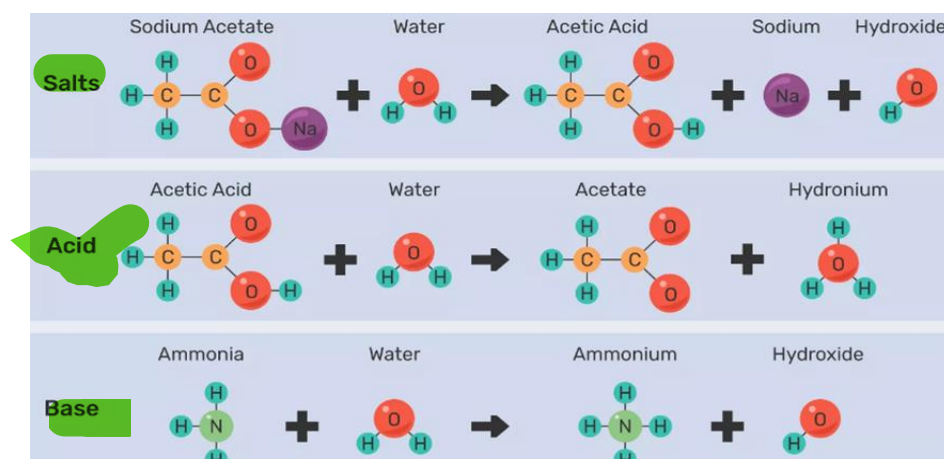


Figure-1: Three common types of hydrolysis reaction.

- Another example is the hydrolysis of ester as shown in the **Figure-2**.

Hydrolysis of ester

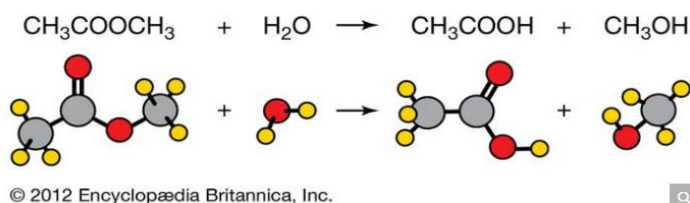


Figure-2: Hydrolysis of methyl acetate to form acetic acid and methyl alcohol.

- There may be thousand of examples where a substance (compounds like starch, proteins, cellulose, hemicellulose, etc.) reacts with water to form or to give simple compounds in order to further the chemical process for achievement of the final objective.

➤ HYDROLYSIS IN BIORESOURCE TECHNOLOGY

- Bioresource technology is an interdisciplinary area which integrates the engineering design and the biological sciences in order to make proper utilization of the bioresources (non-fossil, biogenic origin) to meet the societal needs, such as renewable energy, clean environment and chemicals.
- Amongst the bioresources, biomass is the most prominent source which is composed of the major components: cellulose, hemicellulose and lignin.
- These natural polymeric substances cannot be further converted into usable products, unless or until they are simplified to or converted into simple sugars like glucose or pentose.
- Similarly, starch needs to be converted into sugar, natural fats into fatty acids and glycerol for further applications.

- Here, hydrolysis applied for conversion of starch, natural fats and oils and lignocellulosic materials into simple compounds for further utilization.
- Specially in fermentation, sugars obtained by hydrolysis of starch and lignocellulosic materials are easily converted by the microorganisms into useful products.
- For example: in production of acetone-butanol-ethyl alcohol, using lignocellulosic materials as feedstocks, pretreatment is given through hydrolysis where enzyme, acid, alkali hydrolyses are commonly used.
- Similarly, hydrolysis is commonly used in the production of bio-ATF, bioethanol and other biofuels.
- Hydrolysis may be divided into: physical, chemical, thermochemical, thermophysical, Biological .

➤ PRETREATMENT METHODS FOR LCBs

○ Summary

- ✓ In the past three decades, many studies on the production of biofuels and other chemicals have been conducted using renewable sources such as lignocellulosic biomass.
- ✓ Lignocellulosic biomasses are abundantly available in most countries and furthermore they are carbon neutral.
- ✓ However, the main problem in utilizing lignocellulosic materials lies in the recalcitrance of its bonding.
- ✓ In this lecture, a comprehensive overview and a brief discussion on producing biofuel and valuable chemicals from lignocellulose biomass have been taken up.
- ✓ Various aspects of the physical, chemical, thermophysical, thermochemical and biological pretreatment of lignocellulosic materials have been discussed.
- ✓ The success in biofuel and chemical production strongly depends on the pretreatment method used.
- ✓ Overall, pretreatment is the major step in the successful production of valuable products from lignocellulosic biomass.
- ✓ The various pretreatment methods, the conversion technologies which may be applied after pretreatment, and the primary objectives have been summarized by **Figure-3** and **Figure-4** shown on the next page.

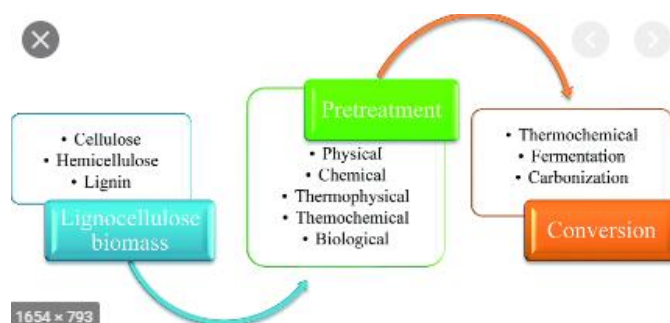


Figure-3: Variou pretreatment methods and the technologies used after treatmeny.

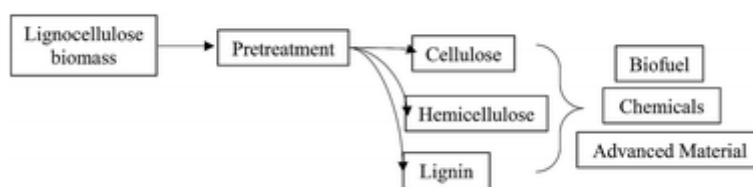


Figure-4: Primary objectives of the pretreatment.

○ Introduction

- ✓ In the past two decades, lignocellulosic biomass (LCB) such as wheat straw, rice straw, sugarcane bagasse, barley, and timothy grass, woody raw materials, forest residues, softwoods and paper pulps have been extensively researched for biofuel.
- ✓ Lignocelluloses are composed of cellulose (40–50%), hemicelluloses (25–35%) and lignin (15– 20%) in an intricate structure where the components are rigidly associated through non-covalent bonds and covalent crosslinkages as shown in **Figure-5(a)**.

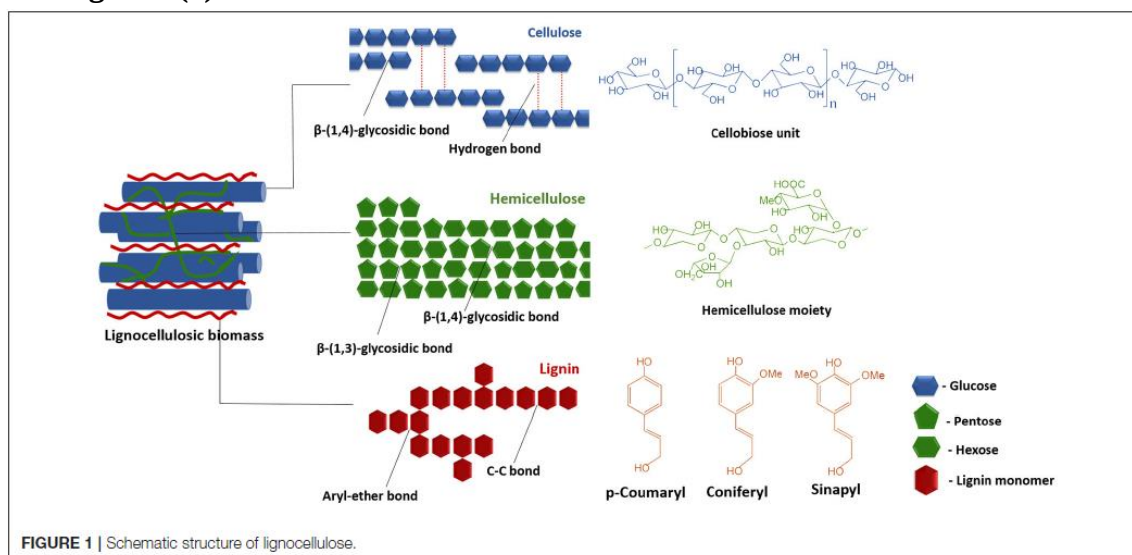


Figure-5 (a): Schematic structure of lignocellulose.

- ✓ Crystallized cellulose and hemicellulose polymer matrix are encrusted by the highly polymerized phenolic lignin that leads to the difficulties in the conversion processes.
- ✓ The complex, hierarchical and recalcitrant nature of the LCB is the primary bottleneck in the utilizing of these resources for bioenergy production or for conversion to any other value added products.
- ✓ This problem has so far been addressed by increasing the digestibility and availability of cellulosic and hemicellulosic fractions, together with the removal of the lignin fractions, through a series of targeted pretreatments
- ✓ The pretreatment techniques currently in use, may be broadly classified as physical, chemical, physicochemical, and biological processes.
- ✓ It is worthwhile to mention that the pretreatment phase contributes to a minimum of 20% of the total cost of conversion for different products and is one area where the energy inputs can be significantly lowered.

- ✓ This urges further study on the understanding of the different techniques and is imperative for the improvement upon the pretreatment phase so as to make the process of conversion of biomass to biofuels and value-added chemicals economical and adaptable to sustainable biorefinery based approaches.
- ✓ In this lecture note, the currently pretreatment techniques for production of biofuels and value-added chemicals, have been discussed with their possible pros and cons.
- ✓ Their recent advancements and effect on different LCBs have also been focussed.
- **Structure of LCB and its components**
 - ✓ LCB is mainly composed of three polymers: cellulose ($C_6H_{10}O_5$)_n, hemicellulose ($C_5H_8O_4$)_m, and lignin [$C_9H_{10}O_3(OCH_3)_{0.9-1.7}$]_x along with minor amounts of other compounds such as proteins, ash, and pectin.
 - ✓ In general, the cellulose, hemicellulose and lignin contents in a typical LCB fall within the range of 30–60, 20–40, and 15–25%, respectively .
 - ✓ However, the composition of these major components varies depending on the source as depicted in **Table-1**.

Table-1: Composition of Various LCBs

lignocellulosic material	cellulose (%)	hemicellulose (%)	lignin (%)
hardwood stems	40–55	24–40	18–25
softwood stems	45–50	25–35	25–35
nut shells	25–30	25–30	30–40
corn cobs	45	35	15
grasses	25–40	35–50	10–30
paper	85–99	0	0–15
wheat straw	30	50	15
sorted refuse	60	20	20
leaves	15–20	80–85	0
cotton seed hairs	80–95	5–20	0
newspaper	40–55	25–40	18–30
waste papers from chemical pulps	60–70	10–20	5–10
primary wastewater solids	8–15		
solid cattle manure	1.6–4.7	1.4–3.3	2.7–5.7
coastal bermudagrass	25	35.7	6.4
switchgrass	45	31.4	12
swine waste	6.0	28	na

- ✓ Cellulose is a main structural and integral part of LCB which is a linear polysaccharide and consists of D-glucose subunits linked by β -(1,4)-glycosidic bonds.
- ✓ This polymer is insoluble in water unless at extremely low or high pH levels.
- ✓ However, it is soluble in solvents like ionic liquids (ILs) and N-methylmorpholine-N-oxide (NMMO).
- ✓ Cellulose possesses the advantageous properties such as biocompatibility, stereoregularity, hydrophilicity, and reactive hydroxyl groups and serves as a versatile resource for derivatized materials such as fibers, films, composites as well as fuels and chemicals.
- ✓ Hemicellulose is a second major component of LCB that consists of short chains of different polysaccharides such as xylan, galactomannan, glucuronoxylan, arabinoxylan, glucomannan, and xyloglucan that are held together by β -(1,4)- and/or β -(1,3)-glycosidic bonds.

- ✓ In contrast to cellulose, hemicellulose is readily degradable into monosaccharides due to low degree of polymerization and non-crystalline nature and thereby widely used in industrial applications such as drug carriers, hydrogels, and cosmetics.
- ✓ Lignin forms a protecting boundary by covalently linking to the cellulose and hemicellulose which enhances the recalcitrance of the lignocellulose.
- ✓ It is a complex, three-dimensional cross-linked polymer that consists of phenyl propane structural units and vary depending on the substitute of the methoxyl groups present in the aromatic rings and are linked to each other by aryl ether linkages e.g., β -O-4, α -O-4 and carbon-carbon bonds e.g., 5-5, β - β .
- ✓ Three basic units that constitute the lignin polymer are p-hydroxyphenyl (H), guaiacyl (G), and syringyl (S).
- ✓ Further, the molecular structure and lignocellulosic composition (cellulose, hemicellulose, and lignin) have been shown in Figure-5 (b).

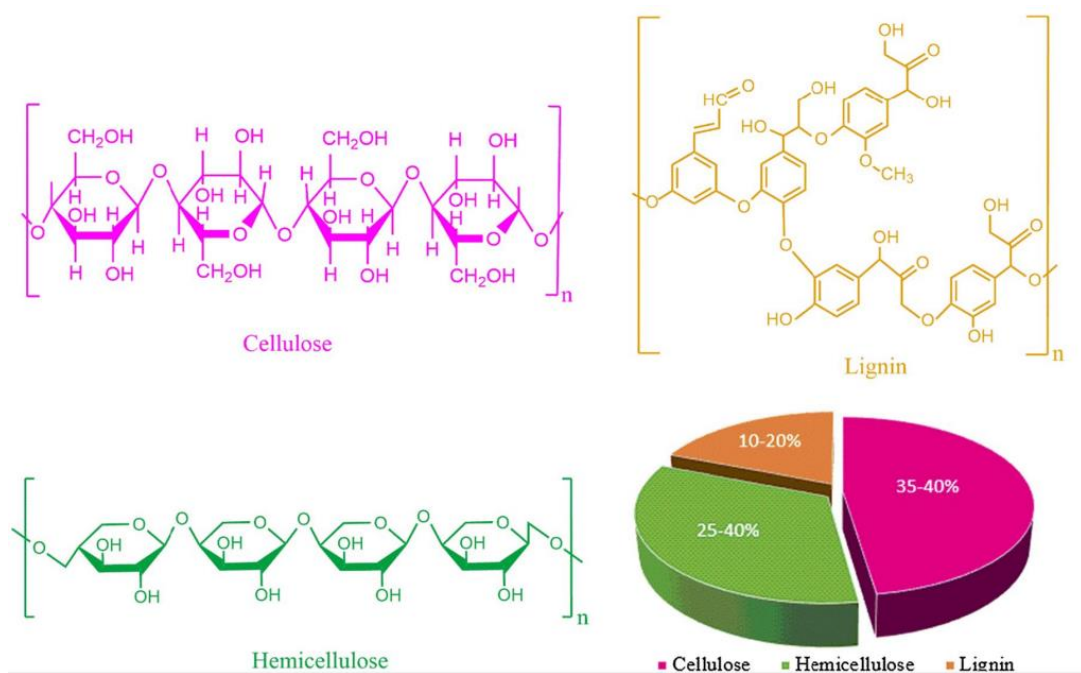


Figure-5 (b): Composition and structure of monomers of cellulose, hemicellulose and lignin.

○ Pretreatment of LCB: Necessity

- ✓ LCBs are resistant to chemical and biological breakdown, known as biomass recalcitrance.
- ✓ Several factors such as the crystalline structure of cellulose, the degree of lignification and the structural heterogeneity and complexity of cell-wall constituents are responsible for the biomass recalcitrance that must be overcome for valuable utilization of lignocellulosic feedstocks.
- ✓ In this context, the pretreatment is a significant step in the biorefinery process.
- ✓ Throughout the pretreatment process, the recalcitrant structure of lignocellulose is disrupted resulting in breakage of lignin sheath, degradation of hemicellulose and reduction in crystallinity and degree of polymerization of cellulose.
- ✓ A multitude of pretreatment techniques have been developed in the last few decades to improve the deconstruction of LCBs and can be categorized as shown in **Figure-6**.

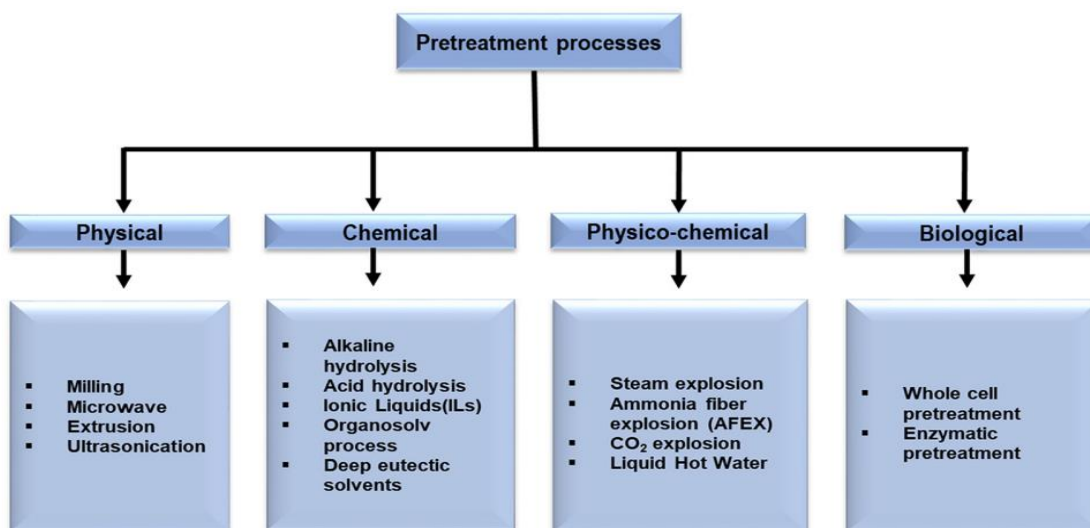


Figure-6: Flow chart diagram of pretreatment processes.

- ✓ Among the several pretreatment processes the preference of the convenient one depends on the type of LCBs used as the composition of cellulose, hemicellulose and lignin vary.
- ✓ In the following sections, the major pretreatment techniques and their effects on the separation of the complex components of various lignocellulosic sources are discussed.
- ✓ Roll of pretreatment has been shown by the **Figure-7**.

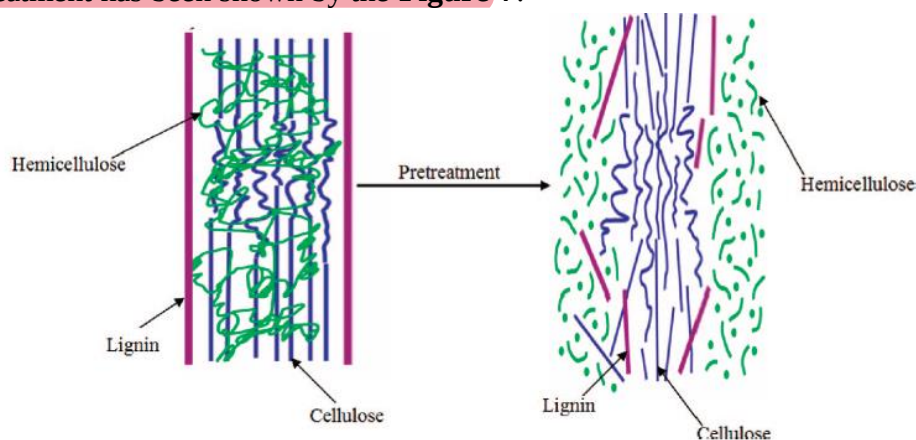


Figure-7: Schematic of the role of pretreatment in the conversion of biomass.

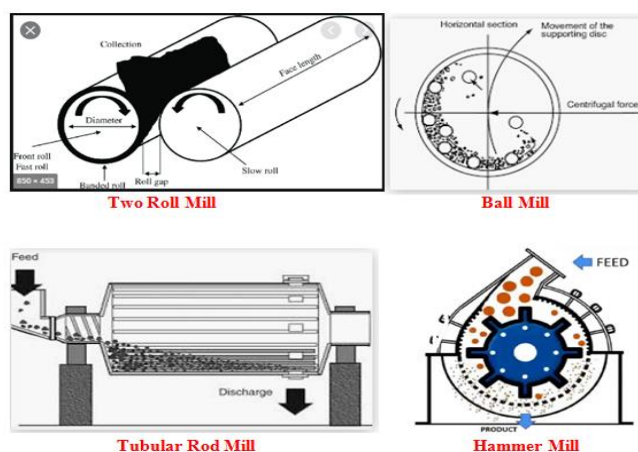
○ **Physical Pretreatment**

- ✓ Physical pretreatment of LCB is a prerequisite prior to any other pretreatment methods.
- ✓ It is primarily carried out to reduce the particle size that results in the increase in surface area, and decrease in degree of polymerization and crystallinity.
- ✓ Consequently, the subsequent processes become more effective and easier.
- ✓ These methods are eco-friendly and seldom produce any toxic material.
- ✓ However, one major disadvantage of physical pretreatment is its high energy consumption.
- ✓ Generally, energy consumption depends on the type of LCB used.

- ✓ It was reported that size reduction of softwoods such as corn stover and switchgrass requires 11.0 and 27.6 kWh/metric ton, respectively, while hardwoods such as pine and poplar chips require 85.4 and 118.5 kWh/metric ton, respectively.
- ✓ The commonly prevalent physical pretreatment methods include milling, extrusion, microwave treatment, and ultrasonication.
- ✓ These approaches are described in detail in the following sections.

Milling

- Milling is employed to reduce the crystallinity and particle size of LCB.
- Milling can reduce the particle size upto 0.2mm.
- However, Chang et al. (1997) unveiled that biomass particle of size <0.4mm has no remarkable effect on the rate and yield of hydrolysis.
- Depending upon the type of motorized equipment operated the different milling methods are two-roll milling, ball milling, rod milling, hammer milling, vibratory milling, colloid milling, and wet disk milling (Figures 8 & 9, below).



Figures-8: Two roll mill, ball mill, tubular rod mill and hammer mill.

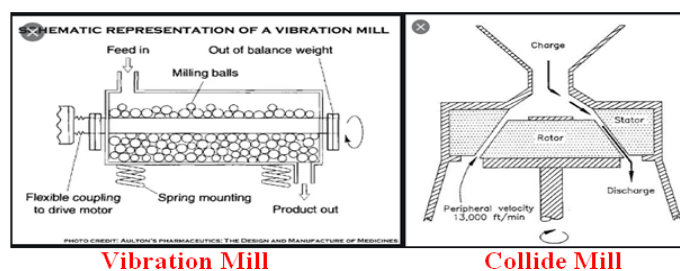


Figure-9: Vibration mill and collide mill.

- The reduction in particle size and crystallinity is determined by the type of milling method adopted, processing time and also the type of biomass used.
- Bai et al. (2018) studied the efficiency of pyrolysis using rod-milling and hammer-milling pretreatment in wheat straw wherein they found significant size reduction and a decrease in crystallinity using rod-milling at an optimum duration of 60 min.
- The effective size reduction and the decrease in crystallinity lead to high surface contact and pore volume of wheat straw.

- Also, the kinetic analysis showed that wheat straw pretreated through rod milling has lower thermal degradation temperature as compared to the hammer-milled wheat straw that enhanced the pyrolysis efficiency.
- However, one of the biggest shortcomings of milling pretreatment is its high energy requirement and the capital cost of mechanical equipment.
- Guet al. (2018) studied the effect of planetary ball milling on premilled wood fiber and found an improved energy consumption efficiency of 0.50–2.15 kWh/kg for 7–30 min of milling at 270 rpm.
- The process produced high glucose and xylose in the range of 24.45–59.67 % and 11.92–23.82 %, respectively through enzymatic hydrolysis.
- Also, mild acid hydrolysis of pretreated cellulose paper and cellulose powder using ball milling resulted in a high yield of nanocellulose with high crystallinity and high thermal stability.
- Milling pretreatment of raw LCB for solid-state anaerobic digestion process showed that reduction in particle size greatly increases the substrate solubility thereby increasing the reaction kinetics.
- Wet diskmilling has been a well-known physical pretreatment owing to its low energy consumption. A study of combined pretreatment using hydrothermal and wet disk milling on oil palm mesocarp fiber (OPMF) demonstrated that hydrothermal pretreatment of OPMF followed by wet disk milling reduces the power consumption upto 9.6 MJ/kg with more than 98% glucose yield.
- Another combined pretreatment of hot compressed water (HCW) and wet disk milling on oil palm biomass produced 85.5 and 100% of total sugar yields from oil palm empty fruit branch and oil palm frond fiber, respectively.
- Kim et al. (2013) compared three different methods of milling; ball, attrition and planetary milling.
- It was found that attrition and planetary milling effectively reduced biomass particle size as compared to ball milling and the highest yield of glucose and galactose was obtained by planetary milling.
- Since milling pretreatment does not result in any toxic or inhibitory compounds it is a preferred preliminary pretreatment method for a wide variety of lignocellulosic feedstocks.

Microwave Assisted Pretreatment

- Microwave irradiation is a non-conventional heating method that has long been used for the pretreatment of LCB under an applied electromagnetic field.
- The first study of microwave irradiation pretreatment was carried out by Ooshima et al.(1984) and since then this method has been retained a convenient one owing to its several advantages including easy operation, energy efficient, minimum inhibitors formation and high heating capacity in short period of time.
- In this method, the dielectric polarization causes molecular collisions and generates thermal energy that results in the disruption of the complex lignocellulosic structure.
- Microwave irradiation was categorized into atmospheric and high-pressure treatment.
- High-pressure microwave pretreatments are operated in closed reactors within the temperature range from 150 to 250°C.

- A study on the microwave pretreatment of *Panicum spp.* and *Miscanthus spp.* showed 7–10% higher solubility of the materials in subcritical water as compared to the untreated materials.
- The samples were pretreated at different temperatures and the optimum conditions obtained are 60 °C and 120 °C for *Miscanthus spp.* and *Panicum spp.*, respectively at 1,600W treatment power.
- In another study, the microwave pretreatment of *Hyacinthus spp.* was examined to enhance the methane production from anaerobic digestion and obtained the highest methane yield of 221 mL/ g-sub which was 38.3% higher than the substrate pretreated with waterheating.
- Nowadays, microwave pretreatment is often added up with other treatment methods as an upgraded attempt.
- Liu et al. (2018) studied the effect of microwave irradiation during alkaline treatment for the separation of cellulose and hemicellulose from a delignified hardwood kraft pulp.
- They found that under microwave treatment the complex fiber structure fractured effectively and the alkaline solution penetrated into the inner fiber structure that resulted in the significant removal of hemicelluloses and high cellulose yield of 93.05%.
- Microwave-assisted ionic liquid treatment of *Crotalaria juncea* fibers produced 78.7 % glucose at 160 °C in only 46 min processing time.
- The studies of the microwave-assisted acid pretreatment of lignin and moso bamboo sawdust have been reported that microwave temperature is the most significant factor in determining the structure of the pretreated samples.
- The microwave temperature facilitates decarboxylation and dehydration during the process.
- Different analysis showed that the pretreated samples have lower O, H, and ash contents and higher C contents, while in raw bamboo sawdust the C, H, and O are predominant.
- Likewise, microwave-assisted acid hydrolysis of jabon kraft pulp produced 49.2 % reducing sugars with 0 % lignin content at 190 °C.
- The lignin removal promotes the production of high sugar yield.

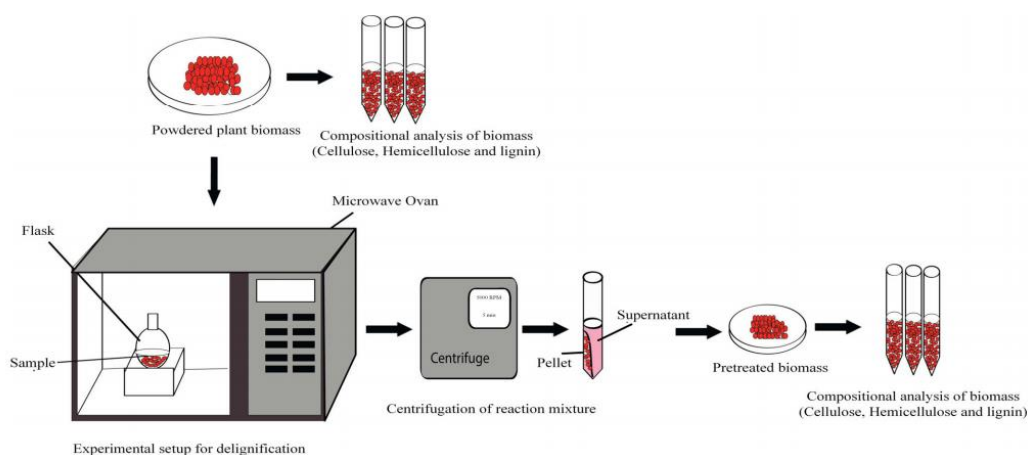


Figure-10: Pictorial representation of experimental setup and process flow of microwave assisted delignification of *Prosopis juliflora* biomass.

- Similarly, Germec et al. (2017) have studied extensively the microwave-assisted dilute acid pretreatment of different agricultural bioresources viz. barley husk, oat husk, wheat bran, and rye bran. The results established the microwave system as an assuring technique for fermentable sugar production from lignocellulosic materials.
- Microwave assisted delignification of *Prosopis juliflora* biomass has been shown in **Figure-10**, below.
- Steam or microwave-assisted sequential salt-alkali pretreatment for lignocellulosic waste (sugarcane leaf waste, SLW) has been shown in **Figure-11**.

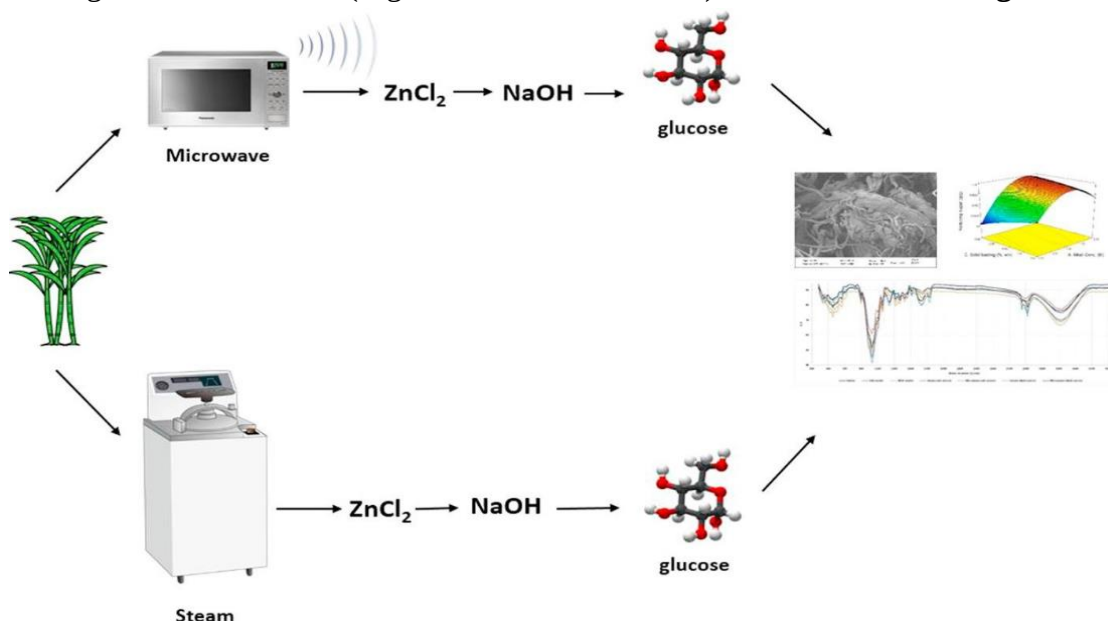


Figure-11: Pictorial representation for microwave assisted delignification of SLW.

Extrusion

- Extrusion is one of the most commonly used physical pretreatment method applied to LCBs.

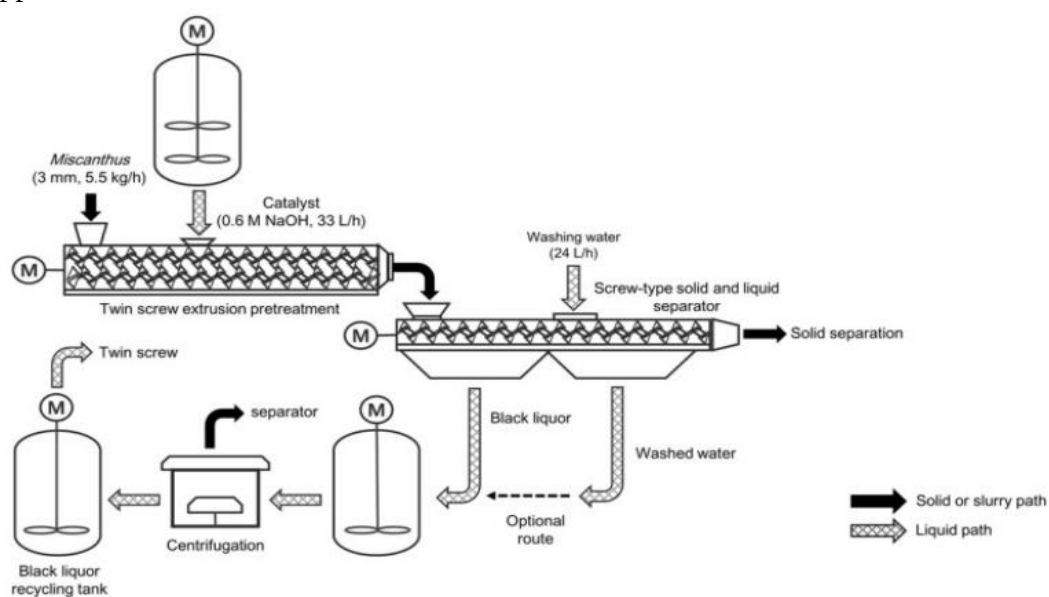


Figure-12: Schematic diagram of the twin crew extrusion pretreatment process for black liquor (BL) recycling.

- The action of one or two screws that spin into a tight barrel, which is furnished with temperature control forms the basis of this method.
- Alkaline twin-screw extrusion pretreatment of *Miscanthus* with recycled black liquor at the pilot scale has been depicted by **Figure-12**.
- The raw materials are passed through the barrel under high temperature ($>300\text{ }^{\circ}\text{C}$) where the recalcitrant structure of the lignocellulose disrupts due to the combined effects of high temperature and the shear forces caused by the rotating screw blades in the barrel.
- Extrusion machines are mainly classified into single screw extruders (made of one single solid piece) and twin screw extruders (made of small pieces called screw elements arranged cylindrically).
- The screw configuration is an important factor affecting the decomposition of LCB.
- Wahid et al. (2015) carried a study on the effect of different screw configurations during extrusion pretreatment on wheat straw and deep litter to enhance biogas production.
- They analyzed pretreatment using five screw configurations namely, mild kneading, long kneading, reverse, kneading, and reverse and kneading with reverse.
- The results showed effective sugar yield with each configurations that consequently led to significant methane production.
- Various parameters such as screw design, screw speed and barrel temperature also control the extrusion pretreatment.
- Moro et al. (2017) studied the pretreatment of sugarcane bagasse and straw using twin-screw extruder and optimized the variables viz. type of additives, biomass:additive ratio, number of extrusion passes, barrel temperature, screw speed, and screw configurations.
- The pretreatment was carried out using different additives such as water, glycerol, ethylene glycol with different loading amounts.
- Experiments showed glycerol as the suitable additive when pretreatment was conducted using a bagasse:glycerol and straw:glycerol ratios of 1:0.75 and 1:0.5 ratios, respectively.

Ultrasonication

- Ultrasonication pretreatment is based on the principle of cavitation through the employment of ultrasonic radiation.
- The cavitation generates shear forces that cleaves the complex network structure of LCB and promotes the extraction of desired compounds such as, cellulose, hemicellulose, and/or lignin.
- A study on the effects of ultrasound pretreatment on the structural changes of eucalyptus wood reveals that the crystallinity of the pretreated wood increased from 34.7 to 35.3% in aqueous soda solution, 32.6– 35.5% in distilled water and 33.4–35.5% in aqueous acetic acid solvent.
- The increased crystallinity is due to the effective removal of the amorphous hemicellulose and lignin fractions that was established by the FTIR analysis.

- It has been seen that the choice of solvents (dilute aqueous solutions of inorganic acids or alkalis, organic solvents or ionic liquids) is critical in determining the optimum conditions for ultrasonication pretreatments.
- Several factors influencing the sonication treatment includes ultrasound frequency, sonication duration, sonication power and temperature.
- Liyakathali et al. (2016) found that the enzymatic digestibility of energy cane bagasse increases with increase in the sonication time and temperature while ultrasonic frequencies had no effect on enzymatic digestibility.
- Cherpozat et al. (2017) studied the use of ultrasonic pretreatment on wood chips for bio-oil production.
- They found the combination of 170 kHz for 0.5 h and 40 kHz for 1.5 h and a power of 1,000W as adequate, resulting 12% increased yield of bio-oil as compared to untreated wood.
- However, ultrasonication for a prolonged period might cause adverse effect due to collision and aggregation between the particles.
- Likewise, high sonication power leads to cavitation near the ultrasound transducer tip that prohibits the energy transfer to the liquid medium.
- Ultrasonication is a viable pretreatment technique owing to its potential to facilitate the disruption of various lignocellulosic materials.
- In fact, the use of ultrasound can scale down the hydrolysis time of biomass up to 80%.

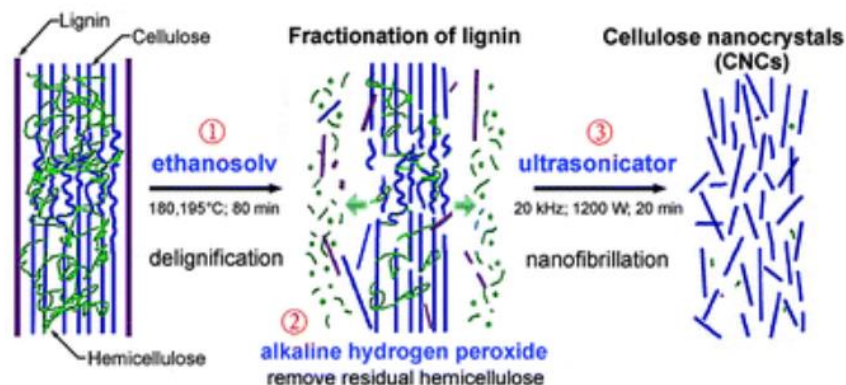


Figure-13: Ultrasonic disintegration of wood flour in two step process.

- However, the process is energy intensive and detailed investigations are necessary to optimize the process parameters for high-scale applications.
- Cellulose nanocrystals (CNCs) were successfully extracted from wood flour by a two-step process that comprised ethanol and peroxide solvothermal pretreatment and an ultrasonic disintegration process as shown in the following **Figure-13..**